Combined Oxidative Phosphorylation Deficiency 58 (COXPD58) — Comprehensive Disease Characteristics Report

Disease: Combined Oxidative Phosphorylation Deficiency 58 (COXPD58) **MONDO:** MONDO:0957537 · **OMIM (phenotype):** 620451 · **Causal gene:** *TEFM* **Report date:** 2026-10-13 · **Evidence base:** small case series + functional/structural studies

Summary (Answer to the Research Question)

Combined Oxidative Phosphorylation Deficiency 58 (COXPD58) is an **ultra-rare autosomal recessive mitochondrial disease** caused by **biallelic loss-of-function/hypomorphic variants in *TEFM*** (Transcription Elongation Factor, Mitochondrial; chr17q11.2). TEFM is the nuclear-encoded elongation factor that increases the processivity of the mitochondrial RNA polymerase POLRMT; its deficiency **reduces promoter-distal mitochondrial RNA transcripts**, lowering synthesis of mtDNA-encoded OXPHOS subunits and producing a **combined (complex I + IV) respiratory-chain deficiency**. Clinically it is a **childhood-onset, variable mitochondrial encephalomyopathy** with near-universal lactic acidosis, hypotonia, intellectual disability, ophthalmologic/ocular-motility signs and seizures, and — in the myopathic subset — a **treatable neuromuscular-junction (NMJ) transmission defect** that can mimic a congenital myasthenic syndrome (Van Haute et al., 2023, *Brain*,).

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Evidence-source note: Content derives from **aggregated disease-level resources** (OMIM/MedGen, HPO, UniProt, gnomAD, ClinVar, HomoloGene) and from **individual-patient primary literature** (a 7-patient defining cohort plus isolated case reports). No large registries, natural-history cohorts, or disease-level omics datasets exist for this very recently defined (2023) entity.

1. Disease Information

Overview. COXPD58 belongs to the clinically and genetically heterogeneous group of "combined oxidative phosphorylation deficiencies" — nuclear-gene mitochondrial disorders in which **more than one respiratory-chain complex is deficient**. COXPD58 specifically results from impaired **mitochondrial transcription elongation**. MedGen/OMIM definition: *"an autosomal recessive disorder characterized by a wide range of clinical presentations including neonatal lactic acidosis, epileptic encephalopathy, developmental delay and impaired intellectual development with nonspecific changes on brain MRI, or mitochondrial myopathy with a treatable neuromuscular transmission defect (Van Haute et al., 2023)."*

Key identifiers. | Resource | ID | |---|---| | MONDO | MONDO:0957537 | | OMIM (phenotype) | 620451 | | OMIM (gene *TEFM*) | 616422 | | UMLS | C5830641 | | MedGen | 1841277 | | GARD | 0026857 | | HGNC (gene) | HGNC:26223 | | NCBI Gene | 79736 | | Ensembl | ENSG00000172171 | | Orphanet | *No dedicated ORPHAcode as of 2026* | | ICD-10 / ICD-11 | No specific code; maps to mitochondrial disease (ICD-10 E88.4x "mitochondrial metabolism disorders"; ICD-11 5C53.1 / 8C70-range) | | MeSH | No specific descriptor; related: "Mitochondrial Diseases" (D028361), "OXPHOS diseases" |

Synonyms / alternative names: COXPD58; Combined oxidative phosphorylation deficiency 58; Combined oxidative phosphorylation defect type 58; *TEFM*-related mitochondrial disease; *TEFM*-associated neurological disorder; mitochondrial transcription elongation factor deficiency.

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2. Etiology

Primary cause — genetic. Biallelic (homozygous or compound-heterozygous) pathogenic variants in ***TEFM***. The disorder is **monogenic and Mendelian (autosomal recessive)**; it is **not infectious or environmentally caused**.

Genetic risk factors. - **Causal variants:** rare biallelic *TEFM* variants (missense, and loss-of-function alleles; large 17q11.2 deletions encompassing *TEFM* are also recorded in ClinVar). - **Reported disease variant:** `NM_024683.4(TEFM):c.469C>G (p.Pro157Ala)` — recurrent, observed in an adolescent male and in two Indian siblings; a possible **population-specific (South Asian) variant** (Adarsha et al., 2025,

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- **Susceptibility/modifier loci:** none established. - **Consanguinity** increases risk of homozygosity (reported in some families), as for most recessive mitochondrial disorders.

Environmental risk factors / protective factors / gene–environment interactions: None identified. As a Mendelian OXPHOS disorder, there are no established environmental triggers, lifestyle risk factors, protective alleles, or GxE interactions. *Inferred, by analogy to other mitochondrial diseases:* physiological stressors (intercurrent infection, fasting, fever, mitochondrial-toxic drugs such as valproate/aminoglycosides) could precipitate metabolic decompensation, but this has not been specifically documented for COXPD58.

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3. Phenotypes

Phenotype frequencies below are from the HPO annotation of OMIM:620451 (source cohort

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denominators reflect the small cohort). **Onset:** childhood in the majority (HP:0011463, 5/7), with neonatal (HP:0003623, 1/7) and infantile (HP:0003593, 1/7) presentations at the severe end. **Severity:** variable (mild treatable myopathy → severe neonatal lactic acidosis/epileptic encephalopathy). **Progression:** chronic; may be progressive (e.g., optic atrophy, spasticity) or relatively stable; episodic decompensations possible.

Laboratory abnormalities (type: laboratory): - **Increased circulating lactate — HP:0002151 — 7/7 (100%)** (hallmark biomarker; CHEBI:16651 L-lactate) - Lacticaciduria — HP:0003648 - Hyperalaninemia — HP:0003348 (CHEBI:16977 L-alanine) - Hypoglycemia — HP:0001943 - Hyperprolinemia — HP:0008358 - Elevated alkaline phosphatase — HP:0003155 - Elevated brain lactate on MR spectroscopy — HP:0012707

Nervous system (signs/symptoms): - Intellectual disability — HP:0001249 — 5/6 - Global developmental delay — HP:0001263; Delayed speech — HP:0000750 - Gait disturbance — HP:0001288 — 4/7; Gait ataxia — HP:0002066 / Ataxia — HP:0001251 - Seizures: Bilateral tonic-clonic — HP:0002069 (2/6); **Epilepsia partialis continua — HP:0012847**; Myoclonus — HP:0001336 - Appendicular spasticity — HP:0034353 (2/6); upper motor neuron signs - CNS demyelination — HP:0007305 (2/2); Cerebellar atrophy — HP:0001272; Thin corpus callosum — HP:0033725; Gliosis — HP:0002171; Microcephaly — HP:0000252 - Exaggerated startle — HP:0002267 - **Peripheral neuropathy — HP:0009830 — 0/7 (notably absent)**

Ophthalmologic (signs): Ptosis — HP:0000508 (3/6); Nystagmus — HP:0000639 (3/6); Strabismus — HP:0000486; **Ophthalmoparesis — HP:0000597** (incl. lateral rectus/abducens palsy); Optic atrophy — HP:0000648 (2/6); Reduced visual acuity/visual loss — HP:0007663/ HP:0000572; Visual fixation instability — HP:0025405.

Musculature (signs + cellular/biopsy): Axial hypotonia — HP:0008936 (5/7); Appendicular hypotonia — HP:0012389; Muscle weakness — HP:0001324; Muscle fatigability (myasthenic-like); **EMG decremental response of CMAP to repetitive nerve stimulation — HP:0003403** (NMJ transmission defect); **Ragged-red fibers — HP:0003200 (2/3)**.

Cellular / biochemical phenotype: Decreased mitochondrial complex I activity — HP:0011923 (3/3); Decreased complex IV activity — HP:0008347 (2/2) → confirms a **combined** OXPHOS defect.

Other: Feeding difficulties — HP:0011968; Vomiting — HP:0002013; Respiratory failure requiring ventilation — HP:0004887 (1/7, severe end); dysmorphic features (epicanthus HP:0000286, low-set ears HP:0000369, prominent nasal bridge) in some.

Quality-of-life impact: not formally measured (no EQ-5D/SF-36/PROMIS data). Expected substantial impact via intellectual disability, motor impairment/ataxia, epilepsy, and visual loss; the treatable NMJ fatigability may improve with therapy.

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4. Genetic / Molecular Information

Causal gene: ***TEFM*** (HGNC:26223; NCBI Gene 79736; Ensembl ENSG00000172171; gene OMIM 616422). Locus **17q11.2**, GRCh38 chr17:30,897,336–30,906,238. Protein: **UniProt Q96QE5**, **360 aa**, with an N-terminal **mitochondrial targeting/transit peptide (aa 1–35)**.

Pathogenic variants. - **Type/class:** missense (e.g., p.Pro157Ala) and loss-of-function alleles; **large contiguous-gene 17q11.2 deletions** spanning *TEFM* are recorded (ClinVar) — e.g., multi-Mb deletions classified Pathogenic. - **Classification:** ClinVar holds **~130 *TEFM* records**, predominantly **VUS** (≈32/50 sampled) and likely benign (5), with **2 Pathogenic + 2 Likely pathogenic** (several "Pathogenic" being large deletions). Variant interpretation remains immature; functional RNA evidence (reduced promoter-distal transcripts) supports pathogenicity. - **Allele frequency / constraint (gnomAD):** *TEFM* is **not LoF-constrained** — pLI = 0.002, observed/expected LoF = 14/22.96 (**oe_lof = 0.61**, 90% CI 0.40–0.95), LOEUF in the tolerant range; missense Z = 0.82. This LoF tolerance explains the **recessive** mechanism (heterozygous carriers are healthy) and the disease's rarity. - **Origin: germline** (not somatic in

this disorder). *(Note: somatically, TEFM is over-expressed in several cancers — see §14 caveat.)* - **Functional consequence: loss of function** (reduced transcription-elongation processivity); no gain-of-function or dominant-negative mechanism described for the disease.

Modifier genes / epigenetics: none established. **Chromosomal abnormalities:** 17q11.2 deletions removing *TEFM* (and neighboring genes, including the *NF1* region) can contribute — relevant for CMA/deletion analysis.

Ontology: gene product GO molecular function **GO:0003711** (transcription elongation factor activity), **GO:0003723** (RNA binding); MONDO:0957537.

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5. Environmental Information

Not applicable. COXPD58 is a monogenic disorder with **no environmental, lifestyle, toxic, or infectious etiology**. No occupational/pollution exposures, dietary factors, or pathogens are implicated. (As a general mitochondrial-disease precaution, avoidance of mitochondrial-toxic agents — e.g., valproate, aminoglycosides, certain antiretrovirals — is prudent but not COXPD58-specific.)

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6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation)

1. **Biallelic *TEFM* variants** reduce the amount or function of the mitochondrial transcription elongation factor (loss of function). *(demonstrated: patient muscle/fibroblasts)*
2. Reduced TEFM **leads to** loss of its "sliding-clamp"/processivity support of POLRMT during mtDNA transcription elongation. *(demonstrated in vitro; Agaronyan 2015, Hillen 2017)*
3. Impaired processivity **results in** premature termination so that **promoter-distal portions of the polycistronic transcripts are under-produced** → reduced promoter-distal mt-RNA. *(demonstrated in patients; Van Haute 2023)*
4. Reduced mt-mRNA/mt-rRNA **leads to** decreased synthesis of **mtDNA-encoded OXPHOS subunits** (complex I ND subunits, complex III cytB, complex IV COI–III, complex V ATP6/8). *(inferred from biochemistry)*

5. This **results in** a **combined respiratory-chain deficiency** (measured: ↓complex I 3/3, ↓complex IV 2/2) and impaired assembly. *(demonstrated)*
6. Deficient OXPHOS **leads to** reduced ATP production and a shift to anaerobic glycolysis → **lactic acidosis** / **hyperalaninemia** (universal biomarker). *(demonstrated)*
7. Bioenergetic failure in **high-energy-demand tissues** (CNS neurons, cerebellum, skeletal muscle, the neuromuscular junction, optic nerve) **results in** the clinical phenotype. *(demonstrated/inferred)*
8. **Branch A (CNS):** neuronal/oligodendrocyte energy failure → developmental delay, intellectual disability, seizures/epilepsia partialis continua, ataxia, CNS demyelination, cerebellar/callosal atrophy.
9. **Branch B (muscle/NMJ):** myopathy with ragged-red fibers **and** a presynaptic/postsynaptic **neuromuscular transmission defect** (decremental RNS) → fatigable weakness, ptosis, ophthalmoparesis — **treatable**, CMS-like.
10. **Branch C (eye):** optic-nerve/retinal energy failure → optic atrophy, reduced acuity.
11. In parallel, because TEFM governs the **transcription↔replication switch** and RNA processing, its loss may also perturb **mtDNA maintenance/copy number and RNA maturation** (Jiang 2019; Matsuda 2025). *(mechanistically inferred for patients)*

Supporting detail

- **Molecular pathway:** mitochondrial gene expression / OXPHOS (Reactome "Mitochondrial transcription"; KEGG oxidative phosphorylation map00190). Upstream machinery: POLRMT, TFAM, TFB2M (initiation) → **TEFM (elongation)**; TEFM is recruited after release of initiation factors (Herbine 2025, P 40712587). Anti-termination at CSBII prevents G-quadruplex/replication-primer formation (Hillen 2017, P 29033127; Agaronyan 2015, P 25635099).
- **Protein dysfunction:** loss of function of the dimeric pseudonuclease-fold elongation factor; structures in PDB 5OL8/5OL9/5OLA, 8U8U/8U8V, 9BDC/9BDD/9MNA.
- **Cellular processes:** energy metabolism failure; secondary oxidative stress possible; no autoimmune/inflammatory mechanism.

- **Metabolic changes:** elevated lactate/pyruvate, alanine, proline; episodic hypoglycemia.
- **Immune involvement:** none.
- **Tissue damage:** bioenergetic/oxidative; gliosis and demyelination documented.
- **Molecular profiling (patients):** transcriptomic signature = **reduced promoter-distal mtRNA** (northern/RNA-seq/qPCR in muscle and fibroblasts); respirometry/enzymology shows combined CI+CIV deficiency.

Ontology suggestions. BP: GO:1903109 (positive regulation of mitochondrial transcription), GO:0006390/GO:0006392 (mitochondrial transcription/elongation), GO:0002082 (regulation of oxidative phosphorylation), GO:0006119 (oxidative phosphorylation), GO:0006264 (mtDNA replication). MF: GO:0003711, GO:0003723. CC: GO:0005759 (mitochondrial matrix), GO:0042645 (mitochondrial nucleoid), GO:1990904 (ribonucleoprotein complex), GO:0005739 (mitochondrion). Cell types (CL): neuron CL:0000540, cerebellar Purkinje cell CL:0000121, skeletal muscle fiber CL:0000188, oligodendrocyte CL:0000128. CHEBI: L-lactate CHEBI:16651, L-alanine CHEBI:16977, ATP CHEBI:30616.

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7. Anatomical Structures Affected

- **Organ level (primary): brain** (UBERON:0000955) — cerebrum, **cerebellum** (UBERON:0002037), **corpus callosum** (UBERON:0002336), white matter; **skeletal muscle** (UBERON:0001134); **eye/optic nerve** (UBERON:0000941). Body systems: **nervous** and **musculoskeletal** predominate; **neuromuscular junction** (UBERON:0002488) specifically involved.
- **Secondary/other:** respiratory (ventilatory failure in severe cases), gastrointestinal (feeding difficulties, vomiting); systemic metabolic (lactic acidosis). No consistent cardiac, hepatic, renal, or endocrine involvement reported (distinguishing from some other COXPDs).
- **Tissue/cell level:** nervous tissue (neurons, oligodendrocytes/myelin), muscle tissue (myofibers with ragged-red change). CL terms: CL:0000540 (neuron), CL:0000128 (oligodendrocyte), CL:0000188 (skeletal muscle fiber), CL:0000121 (Purkinje cell).
- **Subcellular: mitochondrion** — matrix and **nucleoid** (GO:0005759, GO:0042645). This is fundamentally a nucleoid/mitochondrial-matrix gene-expression defect.
- **Localization/lateralization: bilateral/symmetric** CNS and ocular involvement typical; muscle involvement generalized.

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8. Temporal Development

- **Onset:** congenital-to-childhood; **neonatal** (lactic acidosis/encephalopathy) at the severe end, **infantile**, and most commonly **childhood** onset. Pattern: subacute/chronic; may present acutely with neonatal lactic acidosis.
- **Progression:** chronic lifelong; variable rate — some features progressive (optic atrophy, spasticity, white-matter change), others relatively static (developmental). Disease course can include **episodic metabolic decompensation**; epilepsia partialis continua indicates an active encephalopathic process.
- **Patterns:** no spontaneous remission of the underlying disorder; the **NMJ transmission defect may improve with treatment** (treatment-induced symptomatic improvement). **Critical period:** early recognition of the treatable myasthenic component offers a window for symptomatic benefit; early supportive management of lactic acidosis/seizures is important.

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9. Inheritance and Population

- **Epidemiology: ultra-rare.** The literature comprises a **7-patient defining cohort** (Van Haute 2023) plus isolated reports (Adarsha 2025; two Indian siblings) — **well under ~15 reported individuals worldwide.** No prevalence/incidence estimates; **no Orphanet prevalence class.**
- **Inheritance: autosomal recessive** (biallelic). Consistent with gnomAD LoF-tolerance (carriers unaffected).
- **Penetrance:** presumed high/complete in biallelic LoF, but **variable expressivity** is prominent (neonatal-lethal encephalopathy ↔ treatable myopathy). Too few cases for formal penetrance estimates.
- **Anticipation:** not applicable (not a repeat-expansion disorder). **Germline mosaicism:** not reported.
- **Founder effects / population variants: p.Pro157Ala (c.469C>G)** recurs in individuals of **Indian/South Asian** ancestry — a candidate population-specific variant (P 40272035).
- **Consanguinity:** contributes (homozygous variants in consanguineous families).

- **Carrier frequency:** not established; expected very low given rarity. **Sex ratio:** no sex bias expected (autosomal); cohorts too small to assess. **Age distribution:** pediatric predominance.

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10. Diagnostics

Clinical/laboratory tests. - **Blood/CSF lactate** (↑, LOINC 2524-7 lactate; universal), **plasma amino acids** (↑alanine, ↑proline), urine organic acids (lactic aciduria), glucose (episodic hypoglycemia). - **Muscle biopsy:** histochemistry showing **ragged-red fibers**, possible COX-deficient fibers; **respiratory-chain enzymology** demonstrating **combined complex I + IV deficiency** (± others). - **Electrophysiology: repetitive nerve stimulation (RNS)** showing a **decremental CMAP response** (NMJ defect) — a key, treatable clue; EMG may show myopathic features; note RNS can be **normal** in some patients (Adarsha 2025), so a normal RNS does not exclude the disorder. EEG for seizures. - **Imaging:** brain **MRI** — nonspecific changes, white-matter/demyelinating signal, cerebellar atrophy, thin corpus callosum; **MR spectroscopy** — elevated brain lactate.

Genetic testing (confirmatory). - **Whole-exome (WES)** or **whole-genome sequencing (WGS)** is the mainstay — *TEFM* is included on broad **mitochondrial/OXPHOS nuclear-gene panels** and neurometabolic panels; **GeneMatcher** enabled cohort assembly. - **Single-gene / targeted testing** of *TEFM* once suspected (e.g., recurrent p.Pro157Ala in South Asian probands). - **Chromosomal microarray (CMA)/deletion analysis** to detect **17q11.2 deletions** encompassing *TEFM* (and to interpret apparent hemizygosity + a second variant). - **mtDNA testing:** typically **normal** (this is a nuclear-gene disorder) — useful to exclude primary mtDNA disease. - **Functional confirmation:** demonstration of **reduced promoter-distal mtRNA** in fibroblasts/muscle supports VUS reclassification.

Clinical criteria / differential diagnosis. No disease-specific criteria; diagnosis follows **mitochondrial-disease** frameworks plus molecular confirmation. **Differential diagnosis:** other combined OXPHOS deficiencies and mitochondrial translation/transcription defects (e.g., *POLRMT*, *TFAM*, *MTERF*, mt-aminoacyl-tRNA synthetases), **Leigh syndrome**, **congenital myasthenic syndromes** (because of the treatable NMJ defect; distinguished by combined OXPHOS deficiency + lactate), **SLC25A1**-related CMS, and primary mtDNA disorders (MELAS/Leigh). The combination of **lactic acidosis + combined CI/CIV deficiency + a myasthenic RNS decrement** is distinctive.

Screening. Not on newborn-screening panels. **Cascade carrier testing** of relatives and **prenatal/preimplantation testing** are feasible once familial variants are known.

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11. Outcome / Prognosis

- **Survival/mortality:** no formal survival statistics. Prognosis is **severity-dependent**: the **neonatal lactic-acidosis/epileptic-encephalopathy** presentation carries high morbidity/mortality (one patient required assisted ventilation), whereas the **myopathic/NMJ** presentation is comparatively milder and partly treatable.
- **Morbidity/disability:** substantial in encephalopathic patients — intellectual disability, epilepsy, ataxia/spasticity, visual loss; motor fatigability in the myopathic subset.
- **QoL measures:** none formally reported.
- **Complications:** status epilepticus/epilepsia partialis continua, metabolic (lactic) crises, respiratory failure, feeding difficulty/failure to thrive, progressive optic atrophy.
- **Recovery:** the **NMJ transmission defect may respond** to symptomatic treatment; the OXPHOS defect itself is not reversible.
- **Prognostic factors:** age/severity at onset (neonatal worst), degree of respiratory/encephalopathic involvement, and presence of a treatable NMJ component. **Prognostic biomarkers:** lactate levels and residual respiratory-chain enzyme activity (inferred).

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12. Treatment

No curative or disease-specific therapy exists. Management is supportive and, where applicable, targets the treatable NMJ defect.

- **Targeting the NMJ transmission defect (the key actionable feature):** managed as in **congenital myasthenic syndromes — acetylcholinesterase inhibitors (pyridostigmine; NCIT C739)** and consideration of **β2-adrenergic agonists (salbutamol/albuterol; NCIT C28840)** / 3,4-diaminopyridine depending on the synaptic defect. *(Extrapolated from the "treatable neuromuscular transmission defect" description and CMS practice; formal response data in COXPD58 are limited.)*

- **Supportive mitochondrial care (NCIT "Supportive Care" C15277):** antiseizure medications for epilepsy (**avoid valproate** where possible); management of lactic acidosis; nutritional support/feeding assistance; treatment of intercurrent illness; avoidance of mitochondrial-toxic drugs.
- **Rehabilitation:** physical, occupational, and speech therapy (NCIT C15615 Physical Therapy; C15399 Occupational Therapy); low-vision support; developmental/educational support.
- **Empiric "mitochondrial cocktail":** riboflavin (CHEBI/NCIT vitamin B2), **coenzyme Q10** (NCIT C1097), L-carnitine, thiamine, antioxidants — **used empirically, unproven efficacy.**
- **Advanced therapeutics: no** gene therapy, cell therapy, RNA therapy, targeted therapy, or immunotherapy exists or is in trials for COXPD58 (no ClinicalTrials.gov entries identified). Gene replacement is a theoretical future avenue (nuclear-gene, LoF mechanism is conceptually tractable).
- **Pharmacogenomics:** none specific; general mitochondrial-disease drug-avoidance precautions apply.
- **Personalized approach:** recognizing the **treatable myasthenic component** is the principal genotype/phenotype-guided intervention.

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13. Prevention

- **Primary prevention:** not possible (genetic). **Genetic counseling** for recessive 25% recurrence risk; **carrier testing** of at-risk relatives; **prenatal diagnosis / preimplantation genetic testing (PGT-M)** once familial variants are known.
- **Secondary prevention:** early molecular diagnosis (WES/WGS) enables early supportive care and a trial of CMS-type therapy for the NMJ defect; early seizure and metabolic management.
- **Tertiary prevention:** prevent complications — seizure control, nutritional/respiratory support, avoidance of mitochondrial toxins and fasting, low-vision and rehabilitation services.
- **Immunization / public-health / environmental / behavioral measures:** not applicable (non-communicable, non-environmental). Routine childhood vaccination and infection prevention are prudent to avoid metabolic decompensation.

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14. Other Species / Natural Disease

- **Taxonomy / orthologs (HomoloGene 11657):** *TEFM* is **highly conserved across vertebrates** — mouse *Tefm* (NCBI Gene 68550; NCBI:txid10090), rat *Tefm* (287554; txid10116), **zebrafish *tefm* (566236; txid7955)**, *Xenopus tropicalis* (100485913; txid8364), dog (480615; txid9615), cow (534204; txid9913), rhesus macaque (713684; txid9544), chimpanzee (454560; txid9598).
- **Natural disease / breeds (OMIA/VBO): none catalogued** — no naturally occurring *TEFM* disease in companion animals or livestock; no VBO breed associations. Veterinary relevance: nil to date.
- **Comparative biology:** mechanism (mitochondrial transcription elongation) is **evolutionarily conserved**, making cross-species models informative (see §15).
- **Transmission:** not applicable (genetic; **no zoonotic potential**).

15. Model Organisms

- **Zebrafish (*Danio rerio*, txid7955): `tefm` morpholino knockdown** in embryos produced **neuromuscular junction abnormalities and abnormal mitochondrial function**, directly recapitulating the human NMJ/bioenergetic phenotype and strengthening causality (Van Haute 2023,

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  In vivo vertebrate model; good phenotype recapitulation of the NMJ component; limitation: transient knockdown, not a stable genetic model.
- **Mouse (*Mus musculus*, txid10090):** *Tefm* has been studied in vivo/at the cellular level to establish its role in **transcription elongation and RNA processing** (Jiang 2019,

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  , Larsson lab). *Mammalian model of TEFM function; a dedicated COXPD58 disease-model mouse recapitulating the full clinical syndrome is not yet reported; constitutive loss is expected to be severe given essential OXPHOS function.*
- **Cellular / in vitro models: patient-derived primary fibroblasts and skeletal-muscle** showing reduced promoter-distal mtRNA and combined OXPHOS deficiency

(36823193);
cultured cells and reconstituted in vitro transcription systems (POLRMT/TFAM/TFB2M/TEFM) for mechanism and structural studies (PMIDs 25635099, 29033127, 40712587).

- **Resources:** ZFIN (zebrafish), MGI/IMPC (mouse *Tefm*), Cellosaurus (patient lines), PDB (structures).

Important specificity caveat (not disease-relevant but noted): much of the *TEFM* literature concerns its **over-expression as an oncogene** (HCC, glioma, LUAD, endometrial carcinoma; e.g., PMIDs 33771980, 39719635, 39075464, 39731053). These describe **somatic gain**, the *opposite* of the germline loss-of-function that causes COXPD58, and are not part of this Mendelian disease.

Supported vs. Refuted Hypotheses

Supported: 1. COXPD58 is an autosomal recessive disorder caused by biallelic *TEFM* variants (OMIM 620451). ✔ 2. Mechanism = impaired mitochondrial transcription elongation → reduced promoter-distal mtRNA → combined CI/CIV OXPHOS deficiency. ✔ (patient functional data) 3. Phenotype is a variable childhood-onset encephalomyopathy with universal lactic acidosis and a treatable NMJ component. ✔ 4. *TEFM* is LoF-tolerant in gnomAD, consistent with recessive inheritance. ✔ 5. Zebrafish *tefm* knockdown recapitulates the NMJ/mitochondrial phenotype. ✔

Refuted / not supported: - Peripheral neuropathy is **not** a feature (HP:0009830, 0/7). - No environmental, infectious, autoimmune, or somatic/oncogenic mechanism underlies this germline disease. - No dominant-negative/gain-of-function disease mechanism.

Limitations and Future Directions

- **Very small evidence base** (<~15 patients); frequencies derive from a single cohort and are imprecise; no registries, natural-history studies, prevalence estimates, QoL data, or trials.
- **Variant interpretation immature** (mostly VUS in ClinVar); functional assays are needed for reclassification.
- **No approved or trial therapies;** the efficacy of CMS-type drugs for the NMJ defect in COXPD58 needs formal study.

- **Future needs:** stable animal/iPSC-neuron and -myocyte models, longitudinal natural-history data, genotype–phenotype correlation (e.g., p.Pro157Ala), and evaluation of mitochondrial and CMS-directed therapeutics.

Key References (PMIDs)

- **36823193** — Van Haute et al., 2023, *Brain* — defining cohort; *TEFM* variants impair mitochondrial transcription causing childhood-onset neurological disease (reduced promoter-distal mtRNA; zebrafish model; treatable NMJ defect).
- **40272035** — Adarsha et al., 2025 — clinical/radiological characterization; recurrent p.Pro157Ala (South Asian).
- **37239850** — O'Connor et al., 2023 — review: mitochondrial mutations (incl. *TEFM*) and NMJ transmission; proposed "mitochondrial CMS."
- **31036713** — Jiang et al., 2019 — TEFM regulates transcription elongation and RNA processing in mitochondria (in vivo).
- **25635099** — Agaronyan et al., 2015 — TEFM as the replication–transcription switch; processivity.
- **29033127** — Hillen et al., 2017 — structural basis of TEFM anti-termination ("sliding clamp").
- **40712587** / **40236250** — Herbine et al., 2025 — cryo-EM of human mitochondrial transcription; TEFM recruitment after initiation-factor release.
- **39922921** — Matsuda et al., 2025 — TEFM facilitates RNA→DNA synthesis at OriH.
- **21278163** — original identification/characterization of TEFM function (UniProt-cited).
